

PROJECT PROPOSAL

TITLE: MULTIMODAL VISUAL MEDICAL DIAGNOSIS ASSISTANT USING DEEP LEARNING AND RETRIEVAL-AUGMENTED CLINICAL KNOWLEDGE (LAB X)

ABSTRACT

The LAB X Diagnostics Assistant is a privacy-first, local-first clinical decision support system that bridges the gap between deep learning vision diagnostics and peer-reviewed medical literature. This project proposes an intelligent diagnostics portal that leverages convolutional neural networks and Retrieval-Augmented Generation (RAG) to enhance the speed and verifiability of radiological readings. By utilizing local scan uploads, the system automatically classifies pathologies across four distinct modalities—Chest X-Rays, Brain MRIs, Abdominal CTs, and Knee MRIs—and projects explainable attention coordinates (Grad-CAM heatmaps) to outline the diagnostic center. Simultaneously, the system queries an offline FAISS vector database of medical abstracts to synthesize a clinical report with inline PubMed ID citations, eliminating the risk of LLM hallucinations during diagnosis.

OVERVIEW OF THE PROJECT

The project integrates PyTorch deep learning convolutional inference with a responsive Streamlit glassmorphic dashboard UI to provide real-time, explainable diagnostic mappings and cited guidelines retrieval, ensuring that all processing is accelerated locally and no patient data is shared externally.

PROBLEM STATEMENT

During medical evaluations, clinicians and radiologists encounter the following critical challenges:

- **The Explainability Deficit:** Traditional medical deep learning systems operate as 'black boxes', outputting diagnosis codes without visual spatial justification of their classification logic.
- **Clinical Guideline Fragmentation:** Relevant diagnostic standards, PMC articles, and drug dosage rules reside in isolated databases, requiring manual searching that disrupts workflow.
- **EHR Data Privacy Constraints:** Uploading sensitive patient images to open, cloud-hosted LLM endpoints poses severe HIPAA compliance and data leakage risks.
- **Local Hardware Inefficiencies:** Modern neural networks require hardware-level optimizations that can run smoothly on local clinical workstations (MPS/CUDA) without heavy infrastructure budgets.

OBJECTIVES

The primary objectives of the proposed system are as follows:

- **OBJECTIVE 1:** To build a PyTorch convolutional inference pipeline using pre-trained EfficientNet-B4 to classify pathologies across four modalities with high accuracy.
- **OBJECTIVE 2:** To implement a localized Grad-CAM backpropagation layer that generates and overlays 2D Gaussian attention heatmaps at specific anatomical coordinates on scans.
- **OBJECTIVE 3:** To engineer a localized RAG system utilizing an in-memory FAISS vector index to query cited medical abstracts without requiring active cloud network connections.

SCOPE OF THE PROJECT

IN SCOPE (DOABLE TASKS):

- Local processing of medical scans across four modalities (X-Ray, Brain MRI, Abdominal CT, Knee MRI).
- Visual coordinates mapping via Grad-CAM and Matplotlib overlay blending.
- In-memory FAISS vector search of local PubMed journal index modules.
- Formatted diagnostic summary report generation and conversational clinical chatbot.

OUT OF SCOPE (NON-DOABLE TASKS):

- Cloud-based storage of patient radiological scan images.
- Multi-user concurrent DICOM connection queues.
- Integration with physical scanner hardware endpoints via proprietary network protocols.

SYSTEM MODULES/COMPONENTS

1. **Frontend Dashboard UI:** A responsive Streamlit-based interface designed with custom dark-mode radial glassmorphic styling.
2. **Image Preprocessing:** Formats grayscale scan uploads (PIL, Numpy) into standardized 512x512 RGB tensors.
3. **Modality Classifier:** Optimized EfficientNet-B4 PyTorch network structure to run feature classifications.
4. **Grad-CAM Explainability Mapper:** Tracks gradient backpropagation from the last conv layer to identify focal pathology centers.
5. **Literature Indexer & DB:** Text splitter, index generator, and FAISS vector database representing medical abstracts.
6. **RAG Summary Synthesizer:** Translates retrieved vectors and predictions into formatted clinical report markdowns.
7. **Interactive Q&A; Chat Terminal:** Clinician conversational terminal using prompt templates and Google Gemini API keys.
8. **Validation & Verification Suite:** Automated test scripts (test_backend.py) verifying pipeline correctness.

SYSTEM ARCHITECTURE

The system follows a layered architecture:

- **UI Layer:** Built with Streamlit, containing settings sidebar parameters and a double-card visual output grid.
- **Processing Layer:** Handles image matrix resizing, PyTorch MPS/CUDA hardware routing, and Grad-CAM coordinate extraction.
- **Logic Layer:** Compiles vector similarity metrics, queries FAISS nodes, and executes RAG report synthesis.
- **Storage Layer:** Houses local weights (efficientnet_weights.pth), the FAISS index database, and the offline journal collection.

METHODOLOGY

The proposed methodology involves two primary phases:

- 1. **PHASE 1 (Offline Vision Training):** Standardizing datasets to 512x512 pixels, training an EfficientNet-B4 classifier backbone in PyTorch using CrossEntropyLoss, and saving weights.
- 2. **PHASE 2 (Local RAG Integration):** Chunking medical articles, indexing them in FAISS, implementing the Grad-CAM blending, and setting up the Streamlit interface with LLM decoders.

TOOLS AND TECHNOLOGIES

Category	Technology Used
Programming Language	Python 3.9+
User Interface	Streamlit with Custom CSS Injections
Deep Learning Framework	PyTorch, Torchvision
Vector Database	FAISS (Facebook AI Similarity Search)
Clinical Decoders	Google Gemini Pro API / Llama 3 8B
Visualization / Math	Matplotlib, Numpy, Pillow, OpenCV

DATA FLOW

- 1. **Input Data:** Local patient scan image upload + selected modality context.
- 2. **Processing Data:** Vision classifier predicts pathology; Grad-CAM overlays color activation heatmap; FAISS database matches query with medical vectors.
- 3. **Output Data:** Heatmapped diagnostic scan display + cited medical executive report + inline citation chatbot answers.

EXPECTED OUTCOMES

- A responsive, local Streamlit application that successfully maps visual pathologies with Grad-CAM overlays across four scan modalities.
- A secure, hallucination-free decision support system citing valid peer-reviewed clinical abstracts for diagnostic verification.

ADVANTAGES

- **Local Data Privacy:** Processes scans locally without uploading patient data to unencrypted cloud endpoints.
- **Explainable Visual AI:** Shows exactly where the model is looking, giving clinicians high trust indices.
- **Evidence-Based Summaries:** Provides inline PubMed ID citations with diagnostic guidelines.

LIMITATIONS

- **Contrast and Lighting:** Extremely low-contrast scans can throw off accuracy thresholds.
- **API Key Dependencies:** Generating live LLM summary reports requires an active network key.

FUTURE SCOPE

- **PACS Integration:** Connect directly to hospital Picture Archiving systems to load DICOM scans directly.
- **WHO Pathology Expansion:** Extend neural classes to support WHO-grade classifications of tumor sizes and structures.

CONCLUSION

The LAB X Diagnostics Assistant offers a secure, explainable, and highly efficient decision support tool for modern healthcare centers. By bridging the gap between deep learning vision classification and peer-reviewed literature, the system empowers clinicians with visual accuracy and cited research context without compromising patient data security.